

Smart Telescope Sort

User Manual

macOS

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*Organize TIFF and FITS captures from several smart-telescope brands into one year / object library.
This app does not connect to the telescope — you copy sessions onto disk first, choose the matching
telescope type, then Sort.*

Separate from the Vaonis-only **Vespera Sort** program. Demo trees live next to this project under Demo Captures/.

1. Assumptions (read this first)

These assumptions are built into every telescope profile. If your workflow breaks them, Sort will mis-label objects or find nothing.

- **You already moved files off the scope** (USB, FTP, or Wi-Fi). Sort never talks to Vaonis, Seestar, DWARF, or Origin hardware.
- **Source Captures is one brand at a time.** Pick the telescope type in the sidebar to match the folder you point at. Do not mix Seestar albums and Vaonis dated sessions in the same Source Captures root for one preview.
- **Only TIFF / TIF / FIT / FITS are sorted.** JPG previews (Photos, phone album) are ignored. For processing keep TIFF/FITS; JPG is for viewing/sharing only.
- **Apple Photos is not a source.** Export to Files / a Captures folder if images only live in Photos.
- **Destination is always the same layout for every brand:** Targets {year}/{DSO}/ under your Source Captures folder.
- **Year** comes from dated folder names when present; otherwise from the newest image file's modification date.
- **DSO name** is decoded from observation / album / object folder names (normalized to uppercase, spaces → hyphens).
- **Replace only when newer.** If Targets already has the same filename, Sort keeps the newer copy and clears the older source duplicate.
- **Emptied parent capture folders are deleted** after Sort when no TIFF/FITS remain (sidecars go with them). Targets ... folders are never scanned as sources and are never deleted by cleanup.
- **Optional backup** copies whole capture folders before any move.

1.1 What Sort does not do

- It does not run stacking, stretching, or post-processing.
- It does not implement Unistellar (not in this build).
- It does not invent missing object names — if a dump folder has no usable name, the DSO label may be the parent folder name.

2. Where sorted files go (all telescopes)

After Sort, every brand uses the same library shape so multi-night and multi-brand collections stay easy to browse:

```
{Source Captures}/Targets {year}/{DSO}/  
example: .../Captures/Targets 2026/M31/  
example: .../Demo Captures/Seestar/Targets 2026/M42/
```

Parent USB / FTP / dated session folders are emptied as files move. When a parent no longer holds TIFF/FITS, Sort removes that parent at the Captures root.

3. Choose telescope type first

The sidebar **Telescope type** menu selects the scanner. The app remembers your last choice. Then set **Source Captures (...)** to the folder that holds *that* brand's sessions — for demos, the brand subfolder under Demo Captures/, not the parent.

```
/Volumes/Large Drive/Smart Telescope Sort program/Demo Captures/  
Vaonis/      ← telescope type: Vaonis  
Seestar/     ← telescope type: Seestar  
DWARF/       ← telescope type: DWARFLAB
```

```
Origin/      ← telescope type: Origin
README.txt
```

Review file plan opens a sheet listing every planned move / replace / keep (nothing moves yet). **Sort eligible files** offers backup, then moves.

4. How each telescope gets files onto the Mac

Offload is done outside this app. Below: how vendors typically expose files, what Sort expects after you copy them into Captures, and demo coverage.

4.1 Vaonis (Vespera / Stellina)

- **Off the scope:** FTP the Singularity **User/** dated folders (primary). Newer Vespera 3 / Pro 2 also support USB-C transfer. Multi-night: stop each night so the project saves; FTP each night's dated folder.
- **On disk (Captures):** keep dated names unchanged, e.g. 2026-09-21_11-43-55_observation_M31 or ..._plan_My_plan with nested 01-observation-... and 01-images-....
- **Demo:** Demo Captures/Vaonis/ — four sessions (M31×2, NGC7023, plan Demo_Night).

```
User/      (on telescope)
  2026-09-10_22-15-03_observation_M31/
    01-images-initial/
      img-0001.tiff
      img-0002.fits
→ copy whole folder into Captures → Sort → Targets 2026/M31/
```

4.2 ZWO Seestar (S30 / S30 Pro / S50)

- **Off the scope:** USB cable (drive often named Seestar → **MyWorks** / object folders), Wi-Fi / Station Mode file share, or app export of **FIT**. Not FTP-first. JPG often lands in the phone album — that is not Sort's source.
- **On disk (Captures):** object albums with .fit / .fits inside (e.g. M31/, or a USB dump containing several albums).
- **Demo:** Demo Captures/Seestar/ — USB dump, Wi-Fi dump, root M45 album, S50 session.

```
Seestar (USB) / MyWorks/
  M31/
    ...fit
  NGC253/
    ...fits
→ copy albums (or the dump) into Captures → Sort → Targets {year}/M31/
```

4.3 DWARFLAB (DWARF 3 / II / mini)

- **Off the scope:** USB mass storage (appears as a disk) and/or FTP (often ftp://192.168.88.1 on the DWARF Wi-Fi). Session folders hold FITS/TIFF stacks and subs; JPG is for phone viewing.
- **On disk (Captures):** keep each session folder intact. Object may be a nested folder (e.g. .../M33/) under a dated session.
- **Demo:** Demo Captures/DWARF/ — four sessions (DWARF 3, FTP-style, II, mini).

```
DWARF USB / Astronomy / <session>/
  M33/
    stacked.fits
    stacked.tiff
    sub_0001.fits
→ copy session into Captures → Sort → Targets {year}/M33/
```

4.4 Celestron Origin Mark II

- **Off the scope:** enable **Save Raw Images**, then copy via USB stick (FAT32/exFAT) from the app File Manager, or FTP to a computer. Stacked JPG to the phone gallery is not the Sort path.
- **On disk (Captures):** folders named object + date, e.g. M31_2026-09-05/ with .fits (lights; sometimes flats/darks). A parent “USB dump” with several object+date folders is also supported.
- **Demo:** Demo Captures/Origin/ — three object+date folders plus one nested USB dump.

```
M31_2026-09-05/  
  raw_0001.fits    (or Light01.fits on real Origin)  
  raw_0002.fits  
→ copy into Captures → Sort → Targets 2026/M31/
```

5. Recommended workflow

- Copy sessions off the telescope into a Captures-style folder (or use Demo Captures).
- Launch **Smart Telescope Sort**.
- Set **Telescope type** to match those files.
- Set **Source Captures** (...) to that brand's folder.
- Refresh preview. Create Targets {year} if offered.
- Use **Review file plan** to inspect every From → To action.
- Press **Sort eligible files** → optional backup → Sort now.
- Browse Targets {year}/{DSO} for stacking / archive.

6. Demo Captures

Under the project folder, Demo Captures/ holds placeholder TIFF/FITS trees that mimic a post-transfer layout for each brand. They are not real images and not screenshots of vendor Finder windows — they exist so the preview and Sort path behave like a user's machine after USB/FTP copy.

```
/Volumes/Large Drive/Smart Telescope Sort program/Demo Captures/
```

7. Safety rules

- Nothing moves until you confirm Sort (after the backup offer).
- Replace destination files only when the source is newer.
- Never scan Targets {year} as a capture source.
- Never delete the Captures root or any Targets year folder during cleanup.
- Backup copies whole capture trees; it does not modify Targets.

End of manual — Help → Smart Telescope Sort User Manual (■■■) opens this PDF.